

SECTION 1: DOUBLE RIVETED LAP JOINT UNDER TENSION, SHEAR, AND CRUSHING STRESSES

System Parameters: Plate Thickness $t = 15$ mm, Rivet Diameter $d = 25$ mm, Pitch $p = 75$ mm, Ultimate Tensile Stress $\sigma_u = 400$ MPa, Ultimate Shear Stress $\tau_u = 320$ MPa, Ultimate Crushing Stress $\sigma_{cu} = 640$ MPa, Factor of Safety FOS = 4.

Design Protocol: Evaluate plate tearing, rivet shearing ($n = 2$), and rivet crushing strengths to determine minimum rupturing force P_u per pitch; compute working load $P = P_u / \text{FOS}$ and evaluate working stresses.

1. Tearing Strength of Plate (P_t) Plate tearing capacity per pitch length	$P_t = (p - d)t \cdot \sigma_u$$= (75 - 25) \times 15 \times 400$$= 300000 \text{ N}$$\Rightarrow P_t = 300 \text{ kN}$
2. Shearing Strength of Rivets (P_s) Single shear capacity of 2 rivets per pitch length	$P_s = n \cdot \frac{\pi}{4} d^2 \cdot \tau_u$$= 2 \times \frac{\pi}{4} (25)^2 \times 320$$= 314160 \text{ N}$$\Rightarrow P_s = 314.16 \text{ kN}$
3. Crushing Strength of Rivets (P_c) Bearing capacity of 2 rivets per pitch length	$P_c = n \cdot d \cdot t \cdot \sigma_{cu}$$= 2 \times 25 \times 15 \times 640$$= 480000 \text{ N}$$\Rightarrow P_c = 480 \text{ kN}$
4. Rupturing Force per Pitch (P_u) Minimum strength governing joint rupture	$P_u = \min(P_t, P_s, P_c)$$= \min(300, 314.16, 480)$$\Rightarrow P_u = 300 \text{ kN}$

5. Safe Working Load (P) Applied force per pitch at FOS = 4	$P = \frac{P_u}{\text{FOS}}$$= \frac{300}{4}$$= 75 \text{ kN}$$= 75000 \text{ N}$
6. Actual Tensile Stress (σ_t) Induced tensile stress on main plate	$\sigma_t = \frac{P}{(p - d)t}$$= \frac{75000}{(75 - 25) \times 15}$$= 100 \text{ MPa}$
7. Actual Shear Stress (τ) Induced shear stress in rivets	$\tau = \frac{P}{n \cdot \frac{\pi}{4} d^2}$$= \frac{75000}{2 \times \frac{\pi}{4} (25)^2}$$= 76.4 \text{ MPa}$
8. Actual Crushing Stress (σ_c) Induced crushing stress on rivets	$\sigma_c = \frac{P}{n \cdot d \cdot t}$$= \frac{75000}{2 \times 25 \times 15}$$= 100 \text{ MPa}$
Design Output	Rupturing Force P_u = 300 kN Working Load P = 75 kN Working Stresses: σ_t = 100 MPa, τ = 76.4 MPa, σ_c = 100 MPa

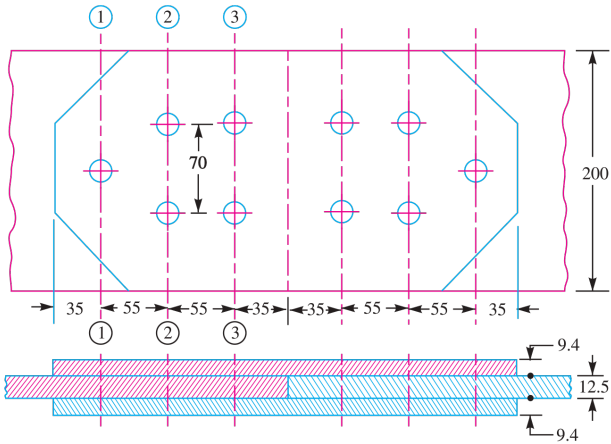


Fig. 9.20. All dimensions in mm.

Fig. 9.20: Double Riveted Lap Joint Scheme

SECTION 2: EFFICIENCY OF SINGLE AND DOUBLE RIVETED LAP JOINTS

System Parameters: Plate Thickness $t = 6$ mm, Rivet Diameter $d = 20$ mm, Case 1 Pitch $p_1 = 50$ mm ($n = 1$), Case 2 Pitch $p_2 = 65$ mm ($n = 2$), Allowable Stresses: Tensile $\sigma_t = 120$ MPa, Shear $\tau = 90$ MPa, Crushing $\sigma_c = 180$ MPa.

Design Protocol: Calculate un-punched solid plate strength, tearing, shearing, and crushing strengths for single and double riveted lap joints; compute joint efficiencies.

1. Solid Plate Strength — Single Lap (P) Strength of un-punched plate ($p = 50$ mm)	$\begin{aligned} P &= p \cdot t \cdot \sigma_t \\ &= 50 \times 6 \times 120 \\ &= 36000 \text{ N} \end{aligned}$
2. Tearing Strength — Single Lap (P_t) Net section tensile capacity	$\begin{aligned} P_t &= (p - d)t \cdot \sigma_t \\ &= (50 - 20) \times 6 \times 120 \\ &= 21600 \text{ N} \end{aligned}$
3. Shearing Strength — Single Lap (P_s) Single shear capacity ($n = 1$)	$\begin{aligned} P_s &= n \cdot \frac{\pi}{4} d^2 \cdot \tau \\ &= 1 \times \frac{\pi}{4} (20)^2 \times 90 \\ &= 28274 \text{ N} \end{aligned}$
4. Crushing Strength — Single Lap (P_c) Bearing capacity ($n = 1$)	$\begin{aligned} P_c &= n \cdot d \cdot t \cdot \sigma_c \\ &= 1 \times 20 \times 6 \times 180 \\ &= 21600 \text{ N} \end{aligned}$

5. Efficiency — Single Riveted Lap Joint (η_1) Ratio of minimum strength to solid strength	$\eta_1 = \frac{\min(P_t, P_s, P_c)}{P} \times 100\%$$= \frac{21600}{36000} \times 100\%$$\Rightarrow \eta_1 = 60.0\%$
6. Solid Plate Strength — Double Lap (P) Strength of un-punched plate ($p = 65$ mm)	$P = p \cdot t \cdot \sigma_t$$= 65 \times 6 \times 120$$= 46800 \text{ N}$
7. Tearing Strength — Double Lap (P_t) Net section tensile capacity	$P_t = (p - d)t \cdot \sigma_t$$= (65 - 20) \times 6 \times 120$$= 32400 \text{ N}$
8. Shearing Strength — Double Lap (P_s) Single shear capacity ($n = 2$)	$P_s = n \cdot \frac{\pi}{4} d^2 \cdot \tau$$= 2 \times \frac{\pi}{4} (20)^2 \times 90$$= 56548 \text{ N}$
9. Crushing Strength — Double Lap (P_c) Bearing capacity ($n = 2$)	$P_c = n \cdot d \cdot t \cdot \sigma_c$$= 2 \times 20 \times 6 \times 180$$= 43200 \text{ N}$
10. Efficiency — Double Riveted Lap Joint (η_2) Ratio of minimum strength to solid strength	$\eta_2 = \frac{\min(P_t, P_s, P_c)}{P} \times 100\%$$= \frac{32400}{46800} \times 100\%$$\Rightarrow \eta_2 = 69.2\%$

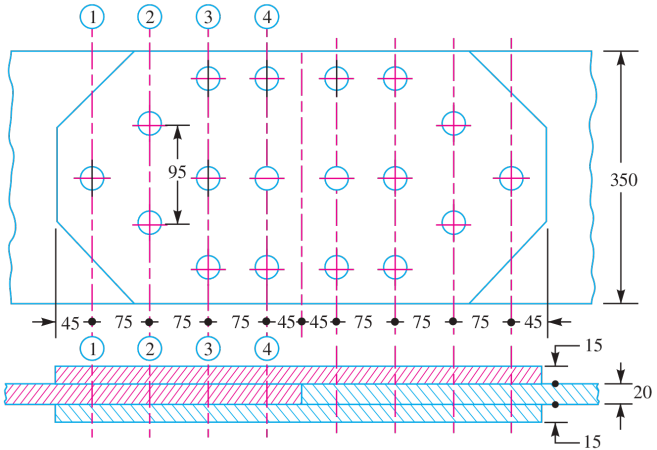


Fig. 9.21. All dimensions in mm.

Fig. 9.21: Single Riveted Lap Joint Scheme

SECTION 3: EFFICIENCY OF DOUBLE RIVETED DOUBLE COVER BUTT JOINT

System Parameters: Plate Thickness $t = 20$ mm, Rivet Diameter $d = 25$ mm, Pitch $p = 100$ mm, Allowable Stresses: Tensile $\sigma_t = 120$ MPa, Shear $\tau = 100$ MPa, Crushing $\sigma_c = 150$ MPa, Double Shear Factor = 2.0.

Design Protocol: Evaluate solid plate strength, plate tearing strength, rivet double shear strength ($n = 2$), and rivet crushing strength to determine joint efficiency.

1. Solid Plate Strength (P) Un-punched solid plate tensile capacity	$P = p \cdot t \cdot \sigma_t$$= 100 \times 20 \times 120$$= 240000 \text{ N}$
2. Tearing Strength of Plate (P_t) Net section tensile strength per pitch	$P_t = (p - d)t \cdot \sigma_t$$= (100 - 25) \times 20 \times 120$$= 180000 \text{ N}$
3. Shearing Strength of Rivets (P_s) Double shear capacity of 2 rivets per pitch ($n = 2$, factor = 2)	$P_s = 2 \cdot \left(2 \cdot \frac{\pi}{4} d^2\right) \cdot \tau$$= 2 \times \left(2 \times \frac{\pi}{4} (25)^2\right) \times 100$$= 196350 \text{ N}$
4. Crushing Strength of Rivets (P_c) Bearing capacity of 2 rivets per pitch ($n = 2$)	$P_c = n \cdot d \cdot t \cdot \sigma_c$$= 2 \times 25 \times 20 \times 150$$= 150000 \text{ N}$

5. Joint Efficiency (η)

Ratio of minimum strength to un-punched solid plate strength

$$\eta = \frac{\min(P_t, P_s, P_c)}{P} \times 100\%$$
$$= \frac{150000}{240000} \times 100\%$$
$$\Rightarrow \eta = 62.5\%$$

Design Output

Double Cover Butt Joint Efficiency $\eta = 62.5\%$ (Governed by Rivet Crushing)

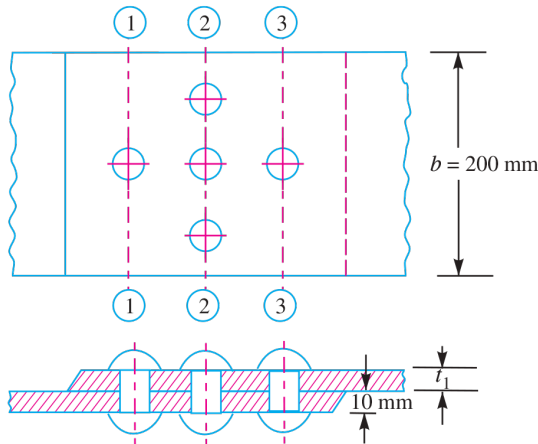


Fig. 9.22

Fig. 9.22: Double Riveted Double Cover Butt Joint

SECTION 4: DOUBLE RIVETED LAP JOINT — ZIG-ZAG RIVETING

System Parameters: Plate Thickness $t = 13\text{ mm}$, Tensile Stress $\sigma_t = 80\text{ MPa}$, Shear Stress $\tau = 60\text{ MPa}$, Crushing Stress $\sigma_c = 120\text{ MPa}$, Rivet Rows $n = 2$ (zig-zag).

Design Protocol: Calculate rivet hole diameter d via Unwin’s formula, determine pitch p by equating $P_t = P_s$ and verify $p \leq p_{\max}$, compute row pitch p_b and margin m , evaluate governing failure mode ($\min(P_t, P_s, P_c)$), and calculate joint efficiency η .

1. Diameter of Rivet Hole (d) Unwin’s formula for $t = 13\text{ mm} > 8\text{ mm}$ and IS standard size	$d = 6\sqrt{t}$ $= 6\sqrt{13} = 21.63\text{ mm}$ $\Rightarrow$ Standard Hole Diameter $d = 23\text{ mm}$ $\Rightarrow$ Nominal Rivet Diameter $d_0 = 22\text{ mm}$
2. Tearing Resistance of Plate (P_t) Net tensile area capacity per pitch length	$P_t = (p - d)t \cdot \sigma_t$ $= (p - 23) \times 13 \times 80$ $= 1040(p - 23)\text{ N}$
3. Shearing Resistance of Rivets (P_s) Single shear capacity for 2 rivets per pitch ($n = 2$)	$P_s = n \cdot \frac{\pi}{4}d^2 \cdot \tau$ $= 2 \times \frac{\pi}{4}(23)^2 \times 60$ $= 2 \times 415.48 \times 60$ $= 49864\text{ N}$
4. Pitch & Max Pitch Limit ($p, p_{\max}$) Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit	$1040(p - 23) = 49864\text{ N}$ $p - 23 = \frac{49864}{1040} = 47.95 \Rightarrow p = 70.95\text{ mm}$ $p_{\max} = C \cdot t + 41.28\text{ mm}$ $= 2.62(13) + 41.28 = 75.34\text{ mm}$ $\Rightarrow$ Adopt Pitch $p = 71\text{ mm}$

5. Distance Between Rows of Rivets (p_b) Transverse row pitch for zig-zag riveting	$p_b = 0.33p + 0.67d$$= 0.33(71) + 0.67(23)$$= 23.43 + 15.41 = 38.84 \text{ mm}$$\Rightarrow \text{Adopt Row Pitch } p_b = 40 \text{ mm}$
6. Margin (m) Distance from rivet hole center to plate edge	$m = 1.5d$$= 1.5(23) = 34.5 \text{ mm}$$\Rightarrow \text{Adopt Margin } m = 35 \text{ mm}$
7. Failure Mode Analysis Tearing (P_t), shearing (P_s), and crushing (P_c) capacities for adopted $p = 71 \text{ mm}$	$P_t = (p - d)t \cdot \sigma_t = (71 - 23)(13)(80) = 49920 \text{ N}$$P_s = n \cdot \frac{\pi}{4}d^2 \cdot \tau = 2\left(\frac{\pi}{4}\right)(23)^2(60) = 49864 \text{ N (Governing)}$$P_c = n \cdot d \cdot t \cdot \sigma_c = 2(23)(13)(120) = 71760 \text{ N}$$\Rightarrow \text{Governing Strength } P_u = P_s = 49864 \text{ N}$$\Rightarrow \text{Failure Mode: Shearing of Rivets}$
8. Solid Plate Strength & Joint Efficiency (P, η) Un-punched solid plate strength and overall joint efficiency	$P = p \cdot t \cdot \sigma_t$$= 71 \times 13 \times 80$$= 73840 \text{ N}$$\eta = \frac{P_s}{P} \times 100\%$$= \frac{49864}{73840} \times 100\%$$\Rightarrow \eta = 67.5\%$
Design Output Summary	Rivet Hole $d = 23 \text{ mm}$ Pitch $p = 71 \text{ mm}$ Row Pitch $p_b = 40 \text{ mm}$ Margin $m = 35 \text{ mm}$ Failure Mode: Shearing Efficiency $\eta = 67.5\%$

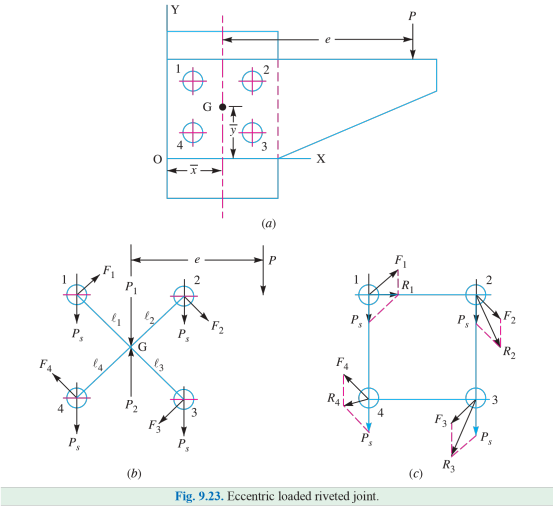


Fig. 9.23. Eccentric loaded riveted joint.

Fig. 9.23: Double Riveted Lap Joint with Zig-Zag Riveting

SECTION 5: TRIPLE RIVETED LAP JOINT — ZIG-ZAG PATTERN

System Parameters: Plate Thickness $t = 7$ mm, Tensile Stress $\sigma_t = 90$ MPa, Shear Stress $\tau = 60$ MPa, Crushing Stress $\sigma_c = 120$ MPa, Rivet Rows $n = 3$ (zig-zag).

Design Protocol: For thin plates ($t < 8$ mm), equate shearing strength to crushing strength ($P_s = P_c$) to determine rivet hole diameter d , equate tearing strength to shearing strength ($P_t = P_s$) to solve for pitch p , limit pitch by I.B.R. maximum pitch ($p \leq p_{\max}$), calculate row pitch p_b , and evaluate governing failure mode.

1. Shearing & Crushing Expressions (P_s, P_c) Symbolic capacities for $t = 7$ mm $<$ 8 mm ($n = 3$ rivets per pitch)	$P_s = n \cdot \frac{\pi}{4} d^2 \cdot \tau = 3 \times \frac{\pi}{4} d^2 \times 60 = 141.4 d^2 \text{ N}$$P_c = n \cdot d \cdot t \cdot \sigma_c = 3 \times d \times 7 \times 120 = 2520 d \text{ N}$
2. Diameter of Rivet Hole (d) Equating shearing strength P_s to crushing strength P_c	$141.4 d^2 = 2520 d$$d = \frac{2520}{141.4} = 17.8 \text{ mm}$ $\Rightarrow$ Standard Hole Diameter $d = 19$ mm $\Rightarrow$ Nominal Rivet Diameter $d_0 = 18$ mm
3. Tearing & Shearing Resistances (P_t, P_s) Capacities evaluated for standard hole diameter $d = 19$ mm	$P_t = (p - d) t \cdot \sigma_t = (p - 19) \times 7 \times 90 = 630 (p - 19) \text{ N}$$P_s = 141.4 d^2 = 141.4 (19)^2 = 51045 \text{ N}$
4. Rivet Pitch & Max Pitch Limit ($p, p_{\max}$) Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit	$630 (p - 19) = 51045 \text{ N}$$p - 19 = \frac{51045}{630} = 81 \Rightarrow p = 100 \text{ mm}$$p_{\max} = C \cdot t + 41.28 \text{ mm}$$= 3.47 (7) + 41.28 = 65.57 \text{ mm}$ $\Rightarrow$ Adopt Pitch $p = p_{\max} = 66$ mm

5. Distance Between Rows of Rivets (p_b)

Transverse row pitch for zig-zag riveting

$$\begin{aligned} p_b &= 0.33p + 0.67d \\ &= 0.33(66) + 0.67(19) \\ &= 21.78 + 12.73 = 34.51 \text{ mm} \\ &\Rightarrow \text{Adopt Row Pitch } p_b = 34.5 \text{ mm} \end{aligned}$$

6. Mode of Failure Analysis

Tearing (P_t), shearing (P_s), and crushing (P_c) capacities for adopted $p = 66 \text{ mm}$

$$\begin{aligned} P_t &= (p - d)t \cdot \sigma_t = (66 - 19)(7)(90) = 29610 \text{ N (Governing)} \\ P_s &= n \cdot \frac{\pi}{4} d^2 \cdot \tau = 3 \left(\frac{\pi}{4} \right) (19)^2 (60) = 51045 \text{ N} \\ P_c &= n \cdot d \cdot t \cdot \sigma_c = 3(19)(7)(120) = 47880 \text{ N} \\ &\Rightarrow \text{Governing Strength } P_u = P_t = 29610 \text{ N} \\ &\Rightarrow \text{Failure Mode: Tearing off the Plate} \end{aligned}$$

Design Output Summary

Rivet Hole $d = 19 \text{ mm}$ | Adopted Pitch $p = 66 \text{ mm}$ | Row Pitch $p_b = 34.5 \text{ mm}$ | Failure Mode: Tearing of Plate

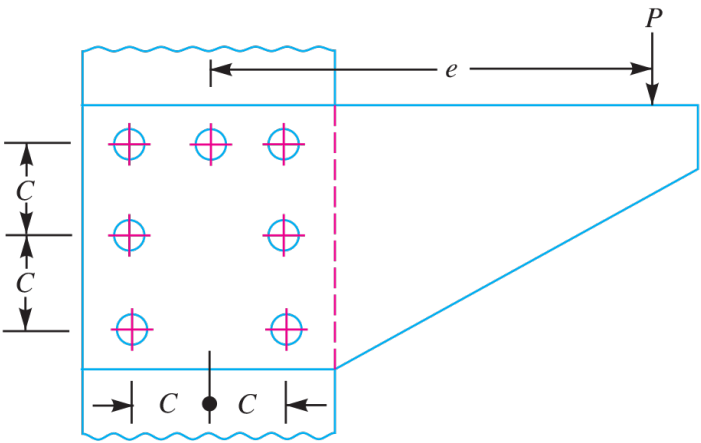


Fig. 9.24

Fig. 9.24: Triple Riveted Lap Joint with Zig-Zag Pattern

SECTION 6: SINGLE RIVETED DOUBLE STRAP BUTT JOINT

System Parameters: Main Plate Thickness $t = 10\text{ mm}$, Tensile Stress $\sigma_t = 80\text{ MPa}$, Shear Stress $\tau = 60\text{ MPa}$, Double Shear Factor = 1.875, Number of Rivets $n = 1$.

Design Protocol: Calculate rivet hole diameter d via Unwin’s formula, determine double shear capacity P_s , state symbolic tearing strength (P_t), solve for pitch p , limit by I.B.R. maximum pitch ($p \leq p_{\max}$), compute cover strap thickness $t_1 = 0.625t$, evaluate governing failure mode, and calculate joint efficiency η .

1. Diameter of Rivet Hole (d) Unwin’s formula for $t = 10\text{ mm} > 8\text{ mm}$ and IS standard size	$d = 6\sqrt{t}$ $= 6\sqrt{10} = 18.97\text{ mm}$ $\Rightarrow$ Standard Hole Diameter $d = 19\text{ mm}$ $\Rightarrow$ Nominal Rivet Diameter $d_0 = 18\text{ mm}$
2. Tearing Resistance of Plate (P_t) Net tensile area capacity per pitch length	$P_t = (p - d)t \cdot \sigma_t$ $= (p - 19) \times 10 \times 80$ $= 800(p - 19)\text{ N}$
3. Double Shear Strength of Rivets (P_s) Double shear capacity for single rivet ($n = 1$, factor 1.875)	$P_s = n \cdot 1.875 \cdot \frac{\pi}{4} d^2 \cdot \tau$ $= 1 \times 1.875 \times \frac{\pi}{4} (19)^2 \times 60$ $= 1.875 \times 283.53 \times 60$ $= \mathbf{31900\text{ N}}$
4. Pitch & Max Pitch Limit ($p, p_{\max}$) Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit	$800(p - 19) = 31900\text{ N}$ $p - 19 = \frac{31900}{800} = 39.87 \Rightarrow p = 58.87\text{ mm}$ $p_{\max} = C \cdot t + 41.28\text{ mm}$ $= 1.75(10) + 41.28 = 58.78\text{ mm}$ $\Rightarrow$ Adopt Pitch $p = p_{\max} = 60\text{ mm}$

5. Cover Strap Thickness (t_1) Standard proportion for equal width double cover straps	$t_1 = 0.625t$ $= 0.625(10)$ $\Rightarrow$ Strap Thickness $t_1 = 6.25$ mm
6. Failure Mode & Joint Efficiency Verification (η) Tearing (P_t), shearing (P_s), solid plate strength (P), and efficiency	$P_t = (p - d)t \cdot \sigma_t = (60 - 19)(10)(80) = 32800$ N $P_s = 31900$ N $\Rightarrow$ Failure Mode: Shearing of Rivets $P = p \cdot t \cdot \sigma_t = 60(10)(80) = 48000$ N $\eta = \frac{P_s}{P} \times 100\%$ $= \frac{31900}{48000} \times 100\%$ $\Rightarrow \eta = 66.5\%$
Design Output Summary	Rivet Hole d = 19 mm Pitch p = 60 mm Cover Strap t1 = 6.25 mm Failure Mode: Shearing Efficiency eta = 66.5%

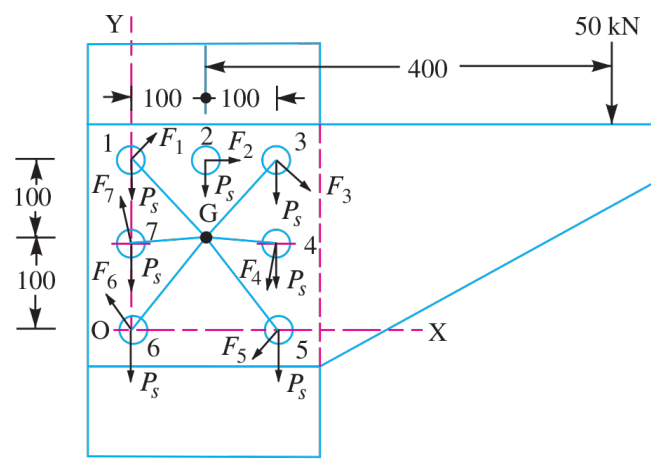


Fig. 9.25

Fig. 9.25: Single Riveted Double Strap Butt Joint

SECTION 7: BOILER LONGITUDINAL DOUBLE RIVETED BUTT JOINT

System Parameters: Boiler Shell Diameter $D = 1.5 \text{ m} = 1500 \text{ mm}$, Steam Pressure $P = 0.95\text{N/mm}^2$, Target Efficiency $\eta_l = 75\%$, Tensile Stress $\sigma_t = 90 \text{ MPa}$, Crushing Stress $\sigma_c = 140 \text{ MPa}$, Shear Stress $\tau = 56 \text{ MPa}$, Double Shear Factor = 1.875, Rivet Rows $n = 2$.

Design Protocol: Calculate shell thickness t , size rivet hole diameter d , state symbolic tearing (P_t) and shearing (P_s) capacities to solve pitch p , limit by I.B.R. max pitch ($p \leq p_{\max}$), calculate row pitch p_b , strap thickness t_1 , and margin m , evaluate failure capacities (P_t, P_s, P_c), and verify joint efficiency η .

1. Boiler Shell Plate Thickness (t)

Thin cylinder formula with 1 mm allowance

$$\begin{aligned} t &= \frac{P \cdot D}{2\sigma_t \cdot \eta_l} + 1 \text{ mm} \\ &= \frac{0.95 \times 1500}{2 \times 90 \times 0.75} + 1 \\ &= \frac{1425}{135} + 1 = 10.55 + 1 = 11.55 \text{ mm} \\ &\Rightarrow \text{Shell Thickness } t = 12 \text{ mm} \end{aligned}$$

2. Diameter of Rivet Hole (d)

Unwin’s formula for $t = 12 \text{ mm} > 8 \text{ mm}$ and IS standard size

$$\begin{aligned} d &= 6\sqrt{t} \\ &= 6\sqrt{12} = 20.78 \text{ mm} \\ &\Rightarrow \text{Standard Hole Diameter } d = 21 \text{ mm} \\ &\Rightarrow \text{Nominal Rivet Diameter } d_0 = 20 \text{ mm} \end{aligned}$$

3. Tearing Resistance of Plate (P_t)

Net tensile area capacity per pitch length

$$\begin{aligned} P_t &= (p - d)t \cdot \sigma_t \\ &= (p - 21) \times 12 \times 90 \\ &= 1080(p - 21) \text{ N} \end{aligned}$$

4. Shearing Resistance of Rivets (P_s)

Double shear capacity for 2 rivets per pitch ($1.875 \times$ single shear)

$$\begin{aligned} P_s &= n \cdot 1.875 \cdot \frac{\pi}{4} d^2 \cdot \tau \\ &= 2 \times 1.875 \times \frac{\pi}{4} (21)^2 \times 56 \\ &= 3.75 \times 346.36 \times 56 \\ &= \mathbf{72745 \text{ N}} \end{aligned}$$

5. Calculated Pitch & I.B.R. Max Pitch Limit ($p, p_{\max}$)

Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit

$$\begin{aligned} 1080(p - 21) &= 72745 \text{ N} \\ p - 21 &= \frac{72745}{1080} = 67.35 \Rightarrow p = 88.35 \text{ mm} \\ p_{\max} &= C \cdot t + 41.28 \text{ mm} \\ &= 3.5(12) + 41.28 = 83.28 \text{ mm} \\ \Rightarrow \text{Adopt Pitch } p &= p_{\max} = \mathbf{84 \text{ mm}} \end{aligned}$$

6. Distance Between Rows of Rivets (p_b)

Transverse row pitch for zig-zag riveting

$$\begin{aligned} p_b &= 0.33p + 0.67d \\ &= 0.33(84) + 0.67(21) \\ &= 27.72 + 14.07 = 41.79 \text{ mm} \\ \Rightarrow \text{Adopt Row Pitch } p_b &= \mathbf{42 \text{ mm}} \end{aligned}$$

7. Thickness of Cover Plates (t_1)

Standard proportion for equal width double cover straps

$$\begin{aligned} t_1 &= 0.625t \\ &= 0.625(12) \\ \Rightarrow \text{Adopt Strap Thickness } t_1 &= \mathbf{7.5 \text{ mm}} \end{aligned}$$

8. Margin (m)

Distance from rivet hole center to plate edge

$$\begin{aligned} m &= 1.5d \\ &= 1.5(21) = 31.5 \text{ mm} \\ \Rightarrow \text{Adopt Margin } m &= \mathbf{32 \text{ mm}} \end{aligned}$$

9. Governing Failure Mode & Joint Efficiency Verification (η)

Tearing capacity P_t , shearing P_s , crushing P_c , solid plate strength P , and efficiency

$$P_t = (84 - 21)(12)(90) = 68040 \text{ N (Governing)}$$

$$P_s = 72745 \text{ N}$$

$$P_c = n \cdot d \cdot t \cdot \sigma_c = 2(21)(12)(140) = 70560 \text{ N}$$

$$P = p \cdot t \cdot \sigma_t = 84(12)(90) = 90720 \text{ N}$$

$$\begin{aligned} \eta &= \frac{P_t}{P} \times 100\% \\ &= \frac{68040}{90720} \times 100\% \\ \Rightarrow \eta &= 75.0\% \quad (\text{Verified matching target efficiency } 75\%) \end{aligned}$$

Design Output Summary

Shell $t = 12 \text{ mm}$ | Hole $d = 21 \text{ mm}$ | Pitch $p = 84 \text{ mm}$ | Row Pitch $p_b = 42 \text{ mm}$ | Strap $t_1 = 7.5 \text{ mm}$ | Margin $m = 32 \text{ mm}$ | Efficiency $\eta = 75\%$

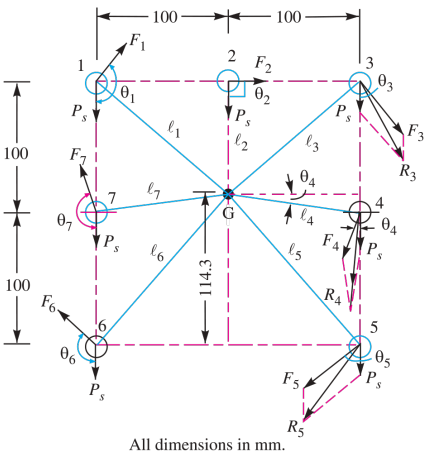


Fig. 9.26

Fig. 9.26: Boiler Longitudinal Double Riveted Double Strap Butt Joint

SECTION 8: PRESSURE VESSEL DOUBLE STRAP BUTT JOINT

System Parameters: Vessel Diameter $D = 1\text{ m} = 1000\text{ mm}$, Internal Pressure $P = 2.75\text{N/mm}^2$, Target Efficiency $\eta_l = 79\%$, Tensile Stress $\sigma_t = 88\text{ MPa}$, Shear Stress $\tau = 64\text{ MPa}$, Double Shear Factor = 1.8, Zig-zag outer pitch $p = 2p_{\text{inner}}$, $n = 3$.

Design Protocol: Calculate shell thickness t , size rivet hole diameter d , state symbolic tearing (P_t) and shearing (P_s) capacities to solve outer pitch p , limit by I.B.R. max pitch ($p \leq p_{\text{max}}$), compute inner pitch p_{inner} , row pitch p_b , strap thickness t_1 , margin m , and evaluate joint efficiency η .

1. Shell Plate Thickness (t)
Thin cylinder formula with 1 mm corrosion/manufacturing allowance

$$t = \frac{P \cdot D}{2\sigma_t \cdot \eta_l} + 1\text{ mm}$$
$$= \frac{2.75 \times 1000}{2 \times 88 \times 0.79} + 1$$
$$= \frac{2750}{139.04} + 1 = 19.78 + 1 = 20.78\text{ mm}$$
$$\Rightarrow \text{Shell Thickness } t = 21\text{ mm}$$

2. Diameter of Rivet Hole (d)
Unwin’s formula for $t = 21\text{ mm} > 8\text{ mm}$ and IS standard size

$$d = 6\sqrt{t}$$
$$= 6\sqrt{21} = 27.5\text{ mm}$$
$$\Rightarrow \text{Standard Hole Diameter } d = 28.5\text{ mm}$$
$$\Rightarrow \text{Nominal Rivet Diameter } d_0 = 27\text{ mm}$$

3. Tearing Resistance of Plate (P_t)
Net tensile area capacity per pitch length in outer row

$$P_t = (p - d)t \cdot \sigma_t$$
$$= (p - 28.5) \times 21 \times 88$$
$$= 1848(p - 28.5)\text{ N}$$

4. Shearing Resistance of Rivets (P_s)

Shear capacity for 3 rivets in double shear ($1.8 \times$ single shear)

$$\begin{aligned} P_s &= n \cdot 1.8 \cdot \frac{\pi}{4} d^2 \cdot \tau \\ &= 3 \times 1.8 \times \frac{\pi}{4} (28.5)^2 \times 64 \\ &= 5.4 \times 637.9 \times 64 \\ &= \mathbf{220500 \text{ N}} \end{aligned}$$

5. Outer & Inner Pitches ($p_{\text{outer}}, p_{\text{inner}}$)

Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit

$$\begin{aligned} 1848(p - 28.5) &= 220500 \text{ N} \\ p - 28.5 &= \frac{220500}{1848} = 119.3 \Rightarrow p = 147.8 \text{ mm} \\ p_{\text{max}} &= C \cdot t + 41.28 \text{ mm} \\ &= 4.63(21) + 41.28 = 138.51 \text{ mm} \\ &\Rightarrow \mathbf{\text{Adopt Outer Pitch } p_{\text{outer}} = p_{\text{max}} = 140 \text{ mm}} \\ &\Rightarrow \mathbf{\text{Adopt Inner Pitch } p_{\text{inner}} = \frac{p}{2} = \frac{140}{2} = 70 \text{ mm}} \end{aligned}$$

6. Distance Between Rows of Rivets (p_b)

I.B.R. row spacing for outer pitch double inner pitch zig-zag butt joint

$$\begin{aligned} p_b &= 0.2p + 1.15d \\ &= 0.2(140) + 1.15(28.5) \\ &= 28 + 32.775 = 60.775 \text{ mm} \\ &\Rightarrow \mathbf{\text{Adopt Row Pitch } p_b = 61 \text{ mm}} \end{aligned}$$

7. Thickness of Butt Straps (t_1)

Equal width double butt straps ratio formula

$$\begin{aligned} t_1 &= 0.625t \cdot \left(\frac{p - d}{p - 2d} \right) \\ &= 0.625(21) \times \left(\frac{140 - 28.5}{140 - 2(28.5)} \right) \\ &= 13.125 \times \left(\frac{111.5}{83} \right) = 17.63 \text{ mm} \\ &\Rightarrow \mathbf{\text{Adopt Strap Thickness } t_1 = 18 \text{ mm}} \end{aligned}$$

8. Margin (<i>m</i>) Distance from rivet hole center to plate edge	$m = 1.5d$ $= 1.5(28.5) = 42.75 \text{ mm}$ $\Rightarrow$ Adopt Margin $m = 43 \text{ mm}$
9. Governing Failure Mode & Joint Efficiency Verification (η) Tearing capacity P_t, solid plate strength P, and efficiency	$P_t = (p - d)t \cdot \sigma_t = (140 - 28.5)(21)(88) = 206050 \text{ N (Governing)}$ $P_s = 220500 \text{ N}$ $P = p \cdot t \cdot \sigma_t = 140(21)(88) = 258720 \text{ N}$ $\eta = \frac{P_t}{P} \times 100\%$ $= \frac{206050}{258720} \times 100\%$ $\Rightarrow \eta = 79.6\% \text{ (Satisfactory, } \geq 79\% \text{)}$
Design Output Summary	Shell t = 21 mm Hole d = 28.5 mm Outer Pitch = 140 mm Inner Pitch = 70 mm Row Pitch pb = 61 mm Strap t1 = 18 mm Margin m = 43 mm Efficiency eta = 79.6%

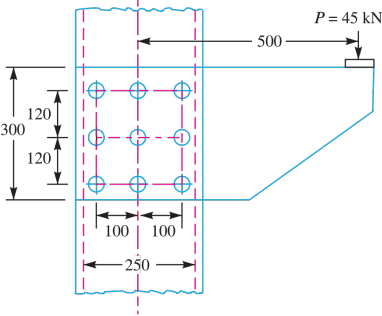


Fig. 9.27

All dimensions in mm.

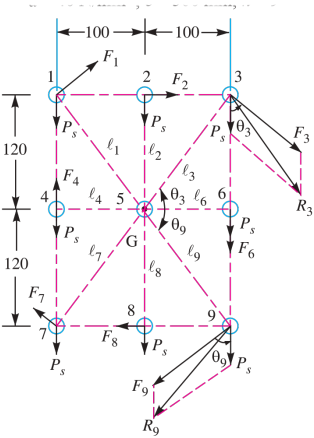


Fig. 9.28

Fig. 9.27 & 9.28: Pressure Vessel Double Strap Butt Joint

SECTION 9: BOILER LONGITUDINAL SEAM DESIGN

System Parameters: Boiler Diameter $D = 1.25\text{ m} = 1250\text{ mm}$, Pressure $P = 2.5\text{N/mm}^2$, Ultimate Tensile Stress $\sigma_{tu} = 420\text{ MPa}$, Ultimate Crushing Stress $\sigma_{cu} = 650\text{ MPa}$, Ultimate Shear Stress $\tau_u = 300\text{ MPa}$, Target Efficiency $\eta_l = 80\%$, Factor of Safety FOS = 5, $n = 5$ rivets per pitch (4 double, 1 single shear).

Design Protocol: Calculate allowable working stresses (σ_t, σ_c, τ), size plate thickness t and rivet hole diameter d , state symbolic tearing (P_t) and shearing (P_s) capacities to solve pitch p , limit by I.B.R. max pitch ($p \leq p_{max}$), compute inner pitch p' , row pitches (p_{b1}, p_{b2}), wide (t_1) and narrow (t_2) strap thicknesses, margin m , and evaluate governing combined failure mode ($P_{t2} + P_{s1}$) to verify joint efficiency η .

1. Allowable Working Stresses

Factor of safety FOS = 5 applied to ultimate material strengths

$$\begin{aligned}\sigma_t &= \frac{\sigma_{tu}}{\text{FOS}} = \frac{420}{5} = 84\text{N/mm}^2 \\ \sigma_c &= \frac{\sigma_{cu}}{\text{FOS}} = \frac{650}{5} = 130\text{N/mm}^2 \\ \tau &= \frac{\tau_u}{\text{FOS}} = \frac{300}{5} = 60\text{N/mm}^2\end{aligned}$$

2. Boiler Shell Plate Thickness (t)

Thin cylinder formula with 1 mm corrosion/manufacturing allowance

$$\begin{aligned}t &= \frac{P \cdot D}{2\sigma_t \cdot \eta_l} + 1\text{ mm} \\ &= \frac{2.5 \times 1250}{2 \times 84 \times 0.80} + 1 \\ &= \frac{3125}{134.4} + 1 = 23.25 + 1 \\ &= 24.25\text{ mm} \\ &\Rightarrow \text{Plate Thickness } t = 25\text{ mm}\end{aligned}$$

3. Diameter of Rivet Hole (d)

Unwin’s formula for $t = 25\text{ mm} > 8\text{ mm}$ and IS standard size

$$\begin{aligned}d &= 6\sqrt{t} \\ &= 6\sqrt{25} = 30\text{ mm} \\ &\Rightarrow \text{Standard Hole Diameter } d = 31.5\text{ mm} \\ &\Rightarrow \text{Nominal Rivet Diameter } d_0 = 30\text{ mm}\end{aligned}$$

4. Tearing Resistance of Plate at Outer Row (P_t)

Net tensile area capacity per pitch length

$$\begin{aligned} P_t &= (p - d)t \cdot \sigma_t \\ &= (p - 31.5) \times 25 \times 84 \\ &= 2100(p - 31.5) \text{ N} \end{aligned}$$

5. Shearing Resistance of Rivets (P_s)

Combined shear capacity for 4 rivets in double shear + 1 rivet in single shear

$$\begin{aligned} P_s &= (4 \cdot 1.875 + 1) \cdot \frac{\pi}{4} d^2 \cdot \tau \\ &= 8.5 \times \frac{\pi}{4} (31.5)^2 \times 60 \\ &= 8.5 \times 779.3 \times 60 \\ &= \mathbf{397500 \text{ N}} \end{aligned}$$

6. Calculated Pitch & I.B.R. Max Pitch Limit ($p, p_{\max}$)

Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit

$$\begin{aligned} 2100(p - 31.5) &= 397500 \text{ N} \\ p - 31.5 &= \frac{397500}{2100} = 189.3 \Rightarrow p = 220.8 \text{ mm} \\ p_{\max} &= C \cdot t + 41.28 \text{ mm} \\ &= 6 \times 25 + 41.28 = 191.28 \text{ mm} \\ &\Rightarrow \mathbf{\text{Adopt Outer Pitch } p = p_{\max} = 196 \text{ mm}} \\ &\Rightarrow \mathbf{\text{Adopt Inner Pitch } p' = \frac{p}{2} = \frac{196}{2} = 98 \text{ mm}} \end{aligned}$$

7. Distance Between Rows of Rivets (p_{b1}, p_{b2})

I.B.R. row spacing for outer-to-next row and inner zig-zag rows

$$\begin{aligned} p_{b1} &= 0.2p + 1.15d \\ &= 0.2(196) + 1.15(31.5) = 39.2 + 36.225 = 75.425 \text{ mm} \\ &\Rightarrow \mathbf{\text{Outer Row Spacing } p_{b1} = 76 \text{ mm}} \\ p_{b2} &= 0.165p + 0.67d \\ &= 0.165(196) + 0.67(31.5) = 32.34 + 21.105 = 53.445 \text{ mm} \\ &\Rightarrow \mathbf{\text{Inner Row Spacing } p_{b2} = 54 \text{ mm}} \end{aligned}$$

8. Butt Strap Thicknesses & Margin (t_1, t_2, m) Inside wide strap, outside narrow strap, and margin	$t_1 = 0.75t = 0.75(25) = 18.75 \text{ mm} \Rightarrow$ Wide Strap $t_1 = 20 \text{ mm}$ $t_2 = 0.625t = 0.625(25) = 15.625 \text{ mm} \Rightarrow$ Narrow Strap $t_2 = 16 \text{ mm}$ $m = 1.5d = 1.5(31.5) = 47.25 \text{ mm} \Rightarrow$ Margin $m = 47.5 \text{ mm}$
9. Individual Failure Mode Capacities Tearing (P_t), shearing (P_s), and crushing (P_c) capacities for $p = 196 \text{ mm}$	$P_{t1} = (p - d)t \cdot \sigma_t = (196 - 31.5)(25)(84) = 345450 \text{ N}$ $P_s = 8.5 \cdot \frac{\pi}{4}d^2 \cdot \tau = 397500 \text{ N}$ $P_c = n \cdot d \cdot t \cdot \sigma_c = 5 \times 31.5 \times 25 \times 130 = 511875 \text{ N}$
10. Combined Row 2 Tearing & Outer Rivet Shear Capacity Tearing of plate at second row combined with shearing 1 outer rivet	$P_{t2} + P_{s1} = (p - 2d)t \cdot \sigma_t + \frac{\pi}{4}d^2 \cdot \tau$ $= (196 - 2 \times 31.5) \times 25 \times 84 + \frac{\pi}{4}(31.5)^2 \times 60$ $= (133)(25)(84) + 47815$ $= 278250 + 47815$ $\Rightarrow$ Governing Least Strength $P_u = 326065 \text{ N}$
11. Solid Plate Strength & Joint Efficiency (P, η) Un-punched solid plate strength and overall joint efficiency	$P = p \cdot t \cdot \sigma_t$ $= 196 \times 25 \times 84$ $= 411600 \text{ N}$ $\eta = \frac{P_u}{P} \times 100\%$ $= \frac{326065}{411600} \times 100\%$ $\Rightarrow \eta = 79.2\% \quad (\text{Satisfactory, nearly equal to target } 80\%)$
Design Output Summary	Shell t = 25 mm Hole d = 31.5 mm Outer Pitch = 196 mm Inner Pitch = 98 mm Row Spacings = 76 mm & 54 mm Straps t1 = 20 mm, t2 = 16 mm Margin m = 47.5 mm Governing Strength = 326065 N Efficiency eta = 79.2%

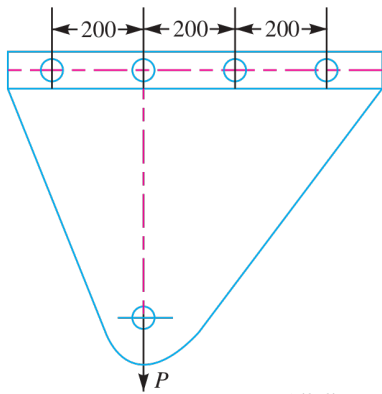


Fig. 9.29

All dimensions in mm.

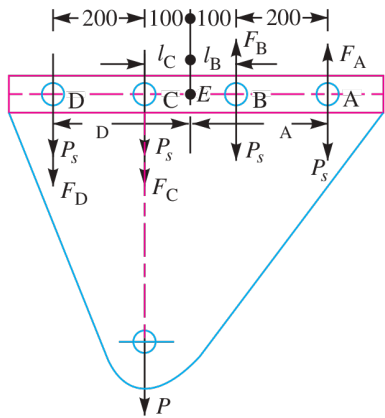


Fig. 9.30

Fig. 9.29 & 9.30: Boiler Longitudinal Seam Triple Riveted Butt Joint

SECTION 10: BOILER LONGITUDINAL & CIRCUMFERENTIAL JOINTS

System Parameters: Boiler Inside Diameter $D = 1.6 \text{ m} = 1600 \text{ mm}$, Working Steam Pressure $P = 2.5\text{N/mm}^2$, Tensile Stress $\sigma_t = 75 \text{ MPa}$, Shear Stress $\tau = 60 \text{ MPa}$, Crushing Stress $\sigma_c = 125 \text{ MPa}$.

Design Protocol: Design Part A (Longitudinal Triple Riveted Butt Joint with unequal straps) for shell thickness t , hole d , pitch p , row spacings, strap thicknesses, and efficiency η_l ; Design Part B (Circumferential Double Riveted Lap Joint) for total steam thrust F , total rivet count n_c , pitch p_1 , row pitch, and efficiency η_c .

1. Shell Plate Thickness (t)
Thin cylinder formula with 1 mm corrosion/manufacturing allowance

$$t = \frac{P \cdot D}{2\sigma_t} + 1 \text{ mm}$$
$$= \frac{2.5 \times 1600}{2 \times 75} + 1$$
$$= \frac{4000}{150} + 1 = 26.67 + 1 = 27.67 \text{ mm}$$
$$\Rightarrow \text{Shell Thickness } t = 28 \text{ mm}$$

2. Diameter of Rivet Hole (d)
Unwin’s formula for $t = 28 \text{ mm} > 8 \text{ mm}$ and IS standard size

$$d = 6\sqrt{t}$$
$$= 6\sqrt{28} = 31.75 \text{ mm}$$
$$\Rightarrow \text{Standard Hole Diameter } d = 34.5 \text{ mm}$$
$$\Rightarrow \text{Nominal Rivet Diameter } d_0 = 33 \text{ mm}$$

3. Longitudinal Tearing & Shearing Capacities (P_t, P_s)
Net section tearing (P_t) and shear capacity for 5 rivets (4 double, 1 single shear)

$$P_t = (p - d)t \cdot \sigma_t = (p - 34.5)(28)(75) = 2100(p - 34.5) \text{ N}$$
$$P_s = (4 \cdot 1.875 + 1) \cdot \frac{\pi}{4}d^2 \cdot \tau = 8.5 \times \frac{\pi}{4}(34.5)^2(60) = 476820 \text{ N}$$

4. Longitudinal Pitch & I.B.R. Max Limit (p, p')

Equating $P_t = P_s$ and enforcing I.B.R. maximum pitch limit ($C = 6$)

$$2100(p - 34.5) = 476820 \text{ N}$$

$$p - 34.5 = \frac{476820}{2100} = 227 \Rightarrow p = 261.5 \text{ mm}$$

$$p_{\text{max}} = C \cdot t + 41.28 = 6(28) + 41.28 = 209.28 \text{ mm}$$

$$\Rightarrow \text{Adopt Outer Pitch } p = 220 \text{ mm}$$

$$\Rightarrow \text{Adopt Inner Pitch } p' = \frac{220}{2} = 110 \text{ mm}$$

5. Longitudinal Row Spacings & Butt Straps ($p_{b1}, p_{b2}, t_1, t_2, m$)

Row pitches, wide inside strap, narrow outside strap, and margin

$$p_{b1} = 0.2p + 1.15d = 0.2(220) + 1.15(34.5) = 83.7 \text{ mm} \Rightarrow \mathbf{85 \text{ mm}}$$

$$p_{b2} = 0.165p + 0.67d = 0.165(220) + 0.67(34.5) = 59.4 \text{ mm} \Rightarrow \mathbf{60 \text{ mm}}$$

$$t_1 = 0.75t = 0.75(28) = \mathbf{21 \text{ mm}}, \quad t_2 = 0.625t = 0.625(28) = \mathbf{18 \text{ mm}}$$

$$m = 1.5d = 1.5(34.5) = 51.75 \text{ mm} \Rightarrow \text{Margin } m = \mathbf{52 \text{ mm}}$$

6. Longitudinal Joint Efficiency Verification (η_l)

Governing combined row 2 tearing + outer rivet shear capacity and efficiency

$$P_{t2} + P_{s1} = (p - 2d)t \cdot \sigma_t + \frac{\pi}{4}d^2 \cdot \tau$$

$$= (220 - 2(34.5))(28)(75) + \frac{\pi}{4}(34.5)^2(60)$$

$$= 317100 + 56096 = \mathbf{373196 \text{ N (Governing Capacity)}}$$

$$P = p \cdot t \cdot \sigma_t = 220(28)(75) = 462000 \text{ N}$$

$$\eta_l = \frac{373196}{462000} \times 100\% \Rightarrow \mathbf{\eta_l = 80.8\%}$$

7. Circumferential Steam End Thrust & Rivet Count (F, n_c)

Total axial force acting on boiler cover and required single shear rivets

$$F = \frac{\pi}{4}D^2 \cdot P = \frac{\pi}{4}(1600)^2(2.5) = 5026548 \text{ N} = 5.027 \text{ MN}$$

$$n_c = \frac{F}{\frac{\pi}{4}d^2 \cdot \tau} = \frac{D^2 P}{d^2 \tau} = (1600)^2 \frac{2.5}{(34.5)^2(60)}$$

$$= \frac{6400000}{71415} = 89.6$$

$$\Rightarrow \text{Total Circumferential Rivets } n_c = \mathbf{90 \text{ rivets}}$$

8. Circumferential Pitch, Row Spacing & Efficiency ($p_1, p_{b,circ}, \eta_c$)

Double riveted lap joint (45 rivets/row), row spacing, margin, and efficiency

$$p_1 = \frac{\pi(D + t)}{45} = \frac{\pi(1600 + 28)}{45} = 113.7 \text{ mm} \Rightarrow \text{Adopt } p_1 = 140 \text{ mm}$$

$$\eta_c = \frac{p_1 - d}{p_1} = \frac{140 - 34.5}{140} \Rightarrow \eta_c = 75.3\%$$

$$p_{b,circ} = 0.33p_1 + 0.67d = 0.33(140) + 0.67(34.5) = 69.3 \text{ mm} \Rightarrow 70 \text{ mm}$$

Design Output Summary

Longitudinal: $t = 28 \text{ mm}$ | Hole $d = 34.5 \text{ mm}$ | Outer Pitch = 220 mm | Straps $t_1 = 21 \text{ mm}$, $t_2 = 18 \text{ mm}$ | $\eta_1 = 80.8\%$ || Circumferential: Thrust $F = 5.027 \text{ MN}$ | Rivets $n_c = 90$ | Pitch $p_1 = 140 \text{ mm}$ | Row Pitch = 70 mm | $\eta_c = 75.3\%$

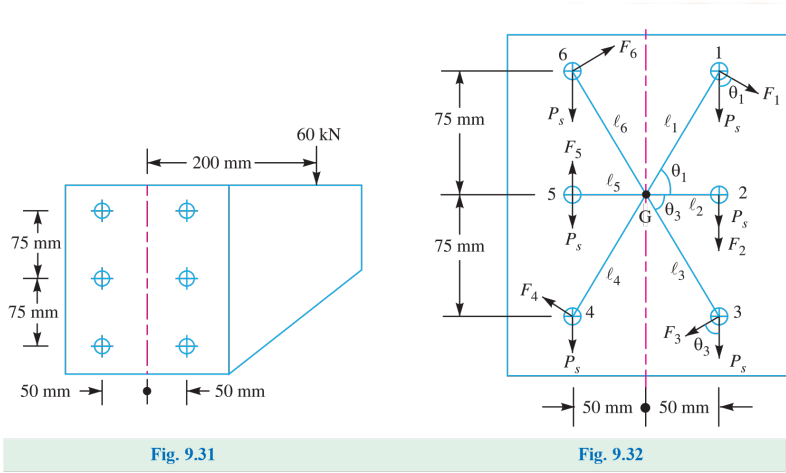


Fig. 9.31 & 9.32: Boiler Longitudinal and Circumferential Joints

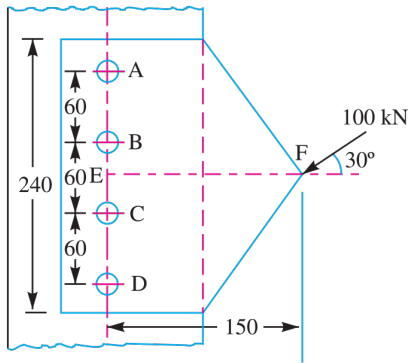
SECTION 11: TIE ROD DOUBLE COVER BUTT JOINT

System Parameters: Tie Rod Width $b = 200$ mm, Thickness $t = 12.5$ mm, Tensile Stress $\sigma_t = 80$ MPa, Shear Stress $\tau = 65$ MPa, Crushing Stress $\sigma_c = 160$ MPa, Double Shear Factor = 1.75.

Design Protocol: Size rivet hole d via Unwin’s formula, determine maximum tensile load P_t at outer section 1-1, compute single rivet shearing (P_{s1}) and crushing (P_{c1}) strengths, solve required rivet count $n = 5$ arranged in diamond pattern (1-2-2), calculate cover strap thickness t_1 , evaluate sectional tearing capacities (P_{t1}, P_{t2}, P_{t3}), verify efficiency $\eta = 89.25\%$, and calculate pitch p , margin m , and row pitch p_b .

1. Diameter of Rivet Hole (d) Unwin’s formula for $t = 12.5$ mm and IS standard size	$d = 6\sqrt{t}$ $= 6\sqrt{12.5} = 21.21$ mm $\Rightarrow$ Standard Hole Diameter $d = 21.5$ mm $\Rightarrow$ Nominal Rivet Diameter $d_0 = 20$ mm
2. Maximum Tensile Load at Section 1-1 (P_t) Net tensile capacity of tie rod at outer row with 1 rivet hole	$P_t = (b - d)t \cdot \sigma_t$ $= (200 - 21.5) \times 12.5 \times 80$ $= 178.5 \times 1000$ $= 178500$ N
3. Single Rivet Capacities & Required Rivets (P_{s1}, P_{c1}, n) Double shear strength ($1.75 \times$ single shear), crushing capacity, and rivet count	$P_{s1} = 1.75 \cdot \frac{\pi}{4}d^2 \cdot \tau = 1.75 \times \frac{\pi}{4}(21.5)^2(65) = 41300$ N $P_{c1} = d \cdot t \cdot \sigma_c = 21.5 \times 12.5 \times 160 = 43000$ N $n = \frac{P_t}{P_{s1}} = \frac{178500}{41300} = 4.32 \Rightarrow n = 5$ rivets (Diamond 1-2-2 Pattern)
4. Thickness of Butt Straps (t_1) Standard proportion for equal width double cover plates	$t_1 = 0.75t$ $= 0.75(12.5) = 9.375$ mm $\Rightarrow$ Strap Thickness $t_1 = 9.4$ mm

5. Sectional Tearing Capacities (P_{t1}, P_{t2}, P_{t3}) Tearing resistances across sections 1-1, 2-2, and 3-3	$P_{t1} = (b - d)t \cdot \sigma_t = (200 - 21.5)(12.5)(80) = 178500 \text{ N}$ (Governing Case) $P_{t2} = (b - 2d)t \cdot \sigma_t + P_{s1} = (200 - 43)(1000) + 41300 = 198300 \text{ N}$ $P_{t3} = (b - 2d)t \cdot \sigma_t + 3P_{s1} = 157000 + 3(41300) = 280900 \text{ N}$
6. Solid Tie Rod Strength & Joint Efficiency (P, η) Strength of un-punched tie rod and overall joint efficiency	$P = b \cdot t \cdot \sigma_t = 200 \times 12.5 \times 80 = 200000 \text{ N}$ $\eta = \frac{P_{t1}}{P} \times 100\% = \frac{178500}{200000} \times 100\%$ $\Rightarrow \eta = 89.25\%$
7. Pitch, Margin & Row Spacing (p, m, p_b) Pitch, margin, and transverse row pitch for diamond layout	$p = 3d + 5 \text{ mm} = 3(21.5) + 5 = 69.5 \text{ mm} \Rightarrow \text{Pitch } p = 70 \text{ mm}$ $m = 1.5d = 1.5(21.5) = 32.25 \text{ mm} \Rightarrow \text{Margin } m = 35 \text{ mm}$ $p_b = 2.5d = 2.5(21.5) = 53.75 \text{ mm} \Rightarrow \text{Row Pitch } p_b = 55 \text{ mm}$
Design Output Summary	Tie Rod b = 200 mm, t = 12.5 mm Hole d = 21.5 mm Rivets n = 5 (1-2-2) Strap t1 = 9.4 mm Pitch p = 70 mm Margin m = 35 mm Row Pitch pb = 55 mm Efficiency eta = 89.25%



All dimensions in mm.

Fig. 9.33

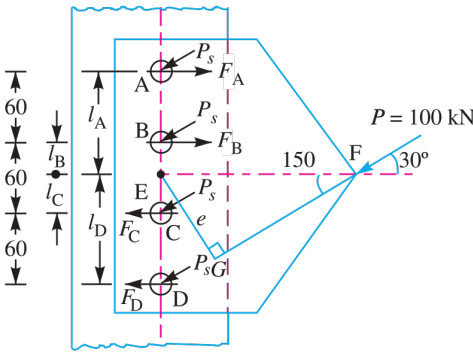


Fig. 9.34

Fig. 9.33 & 9.34: Tie Rod Double Cover Butt Joint

SECTION 12: BRIDGE TIE-BAR LOZENGE (DIAMOND) JOINT

System Parameters: Flat Bar Width $b = 350$ mm, Thickness $t = 20$ mm, Tensile Stress $\sigma_t = 90$ MPa, Shear Stress $\tau = 60$ MPa, Crushing Stress $\sigma_c = 150$ MPa, Double Shear Factor = 1.75.

Design Protocol: Size rivet hole diameter d via Unwin’s formula, determine maximum tensile load P_t at outer section 1-1, compute single rivet shearing (P_{s1}) and crushing (P_{c1}) strengths, solve required rivet count $n = 9$ arranged in lozenge pattern (1-2-3-3), calculate cover strap thickness $t_1 = 15$ mm, evaluate sectional tearing capacities ($P_{t1}, P_{t2}, P_{t3}, P_{t4}$), verify efficiency $\eta = 91.7\%$, and compute pitch $p = 95$ mm, margin $m = 45$ mm, and row pitch $p_b = 75$ mm.

1. Diameter of Rivet Hole (d) Unwin’s formula for $t = 20$ mm and IS standard size	$d = 6\sqrt{t}$ $= 6\sqrt{20} = 26.83$ mm $\Rightarrow$ Standard Hole Diameter $d = 29$ mm $\Rightarrow$ Nominal Rivet Diameter $d_0 = 27$ mm
2. Maximum Tensile Load at Section 1-1 (P_t) Net tensile capacity of tie-bar at outer row with 1 rivet hole	$P_t = (b - d)t \cdot \sigma_t$ $= (350 - 29) \times 20 \times 90$ $= 321 \times 1800$ $= 577800$ N
3. Single Rivet Capacities & Required Rivets (P_{s1}, P_{c1}, n) Double shear strength ($1.75 \times$ single shear), crushing capacity, and total rivet count	$P_{s1} = 1.75 \cdot \frac{\pi}{4}d^2 \cdot \tau = 1.75 \times \frac{\pi}{4}(29)^2(60) = 69360$ N $P_{c1} = d \cdot t \cdot \sigma_c = 29 \times 20 \times 150 = 87000$ N $n = \frac{P_t}{P_{s1}} = \frac{577800}{69360} = 8.33 \Rightarrow n = 9$ rivets (Lozenge 1-2-3-3 Pattern)
4. Thickness of Butt Straps (t_1) Standard proportion for equal width double cover plates	$t_1 = 0.75t$ $= 0.75(20)$ $\Rightarrow$ Adopt Strap Thickness $t_1 = 15$ mm

5. Sectional Tearing Capacities ($P_{t1}, P_{t2}, P_{t3}, P_{t4}$) Tearing resistances across sections 1-1 (1 hole), 2-2 (2 holes), 3-3 (3 holes), and 4-4 (3 holes)	$P_{t1} = (b - d)t \cdot \sigma_t = (350 - 29)(20)(90) = 577800 \text{ N}$ (Governing Capacity) $P_{t2} = (b - 2d)t \cdot \sigma_t + P_{s1} = (350 - 58)(1800) + 69360 = 594960 \text{ N}$ $P_{t3} = (b - 3d)t \cdot \sigma_t + 3P_{s1} = (350 - 87)(1800) + 208080 = 681480 \text{ N}$ $P_{t4} = (b - 3d)t \cdot \sigma_t + 6P_{s1} = 473400 + 416160 = 889560 \text{ N}$
6. Solid Flat Bar Strength & Joint Efficiency (P, η) Strength of un-punched tie-bar and overall diamond joint efficiency	$P = b \cdot t \cdot \sigma_t = 350 \times 20 \times 90 = 630000 \text{ N}$ $\eta = \frac{P_{t1}}{P} \times 100\% = \frac{577800}{630000} \times 100\%$ $\Rightarrow \eta = 91.7\%$ (Superior to Chain Riveting 75.2%)
7. Pitch, Margin & Row Spacing (p, m, p_b) Pitch, margin, and transverse row pitch for lozenge layout	$p = 3d + 5 \text{ mm} = 3(29) + 5 = 92 \text{ mm} \Rightarrow \text{Pitch } p = 95 \text{ mm}$ $m = 1.5d = 1.5(29) = 43.5 \text{ mm} \Rightarrow \text{Margin } m = 45 \text{ mm}$ $p_b = 2.5d = 2.5(29) = 72.5 \text{ mm} \Rightarrow \text{Row Pitch } p_b = 75 \text{ mm}$
Design Output Summary	Tie-Bar $b = 350 \text{ mm}$, $t = 20 \text{ mm}$ Hole $d = 29 \text{ mm}$ Rivets $n = 9$ (1-2-3-3) Strap $t1 = 15 \text{ mm}$ Pitch $p = 95 \text{ mm}$ Margin $m = 45 \text{ mm}$ Row Pitch $p_b = 75 \text{ mm}$ Efficiency $\eta = 91.7\%$

SECTION 13: FLAT TIE-BAR LAP JOINT (DIAMOND PATTERN)

System Parameters: Flat Bar Width $b = 200$ mm, Thickness $t = 10$ mm, Nominal Rivet Diameter $d_0 = 24$ mm, Rivet Hole Diameter $d = 25.5$ mm, Allowable Stresses: Tensile $\sigma_t = 112$ MPa, Shear $\tau = 84$ MPa, Crushing $\sigma_c = 200$ MPa.

Design Protocol: Calculate maximum pull P_t at outer section 1-1, determine single rivet shearing (P_{s1}) and crushing (P_{c1}) strengths, solve required rivet count $n = 5$ arranged in diamond pattern (1-3-1), calculate cover plate thickness $t_1 = 1.25t = 12.5$ mm, evaluate sectional tearing capacities (P_{t1}, P_{t2}, P_{t3}), and verify governing joint efficiency $\eta = 80.9\%$.

1. Maximum Tensile Load at Section 1-1 (P_t) Net tensile capacity of tie-bar at outer row with 1 rivet hole ($d = 25.5$ mm)	$P_t = (b - d)t \cdot \sigma_t$$= (200 - 25.5) \times 10 \times 112$$= 174.5 \times 1120$$= \mathbf{195440\text{ N}}$
2. Single Rivet Capacities & Required Rivets (P_{s1}, P_{c1}, n) Single shear strength in lap joint, crushing capacity, and total rivet count	$P_{s1} = \frac{\pi}{4}d^2 \cdot \tau = \frac{\pi}{4}(25.5)^2(84) = 42905\text{ N (Governing)}$$P_{c1} = d \cdot t \cdot \sigma_c = 25.5 \times 10 \times 200 = 51000\text{ N}$$n = \frac{P_t}{P_{s1}} = \frac{195440}{42905} = 4.56 \Rightarrow \mathbf{n = 5\text{ rivets (Diamond 1-3-1 Pattern)}}$
3. Thickness of Cover Plate (t_1) Standard proportion for lap joint cover plate	$t_1 = 1.25t$$= 1.25(10)$$\Rightarrow \mathbf{\text{Cover Plate Thickness } t_1 = 12.5\text{ mm}}$
4. Sectional Tearing Capacities (P_{t1}, P_{t2}, P_{t3}) Tearing resistances across sections 1-1 (1 hole), 2-2 (3 holes), and 3-3 (1 hole)	$P_{t1} = (b - d)t \cdot \sigma_t = (200 - 25.5)(10)(112) = 195440\text{ N}$$P_{t2} = (b - 3d)t \cdot \sigma_t + P_{s1} = (200 - 3(25.5))(1120) + 42905$$= 138320 + 42905 = \mathbf{181225\text{ N (Governing Minimum Strength)}}$$P_{t3} = (b - d)t \cdot \sigma_t = 195440\text{ N}$

SECTION 14: ECCENTRICALLY LOADED LAP RIVETED BRACKET

System Parameters: Bracket Thickness $t = 25$ mm, Eccentric Load $P = 50$ kN = 50000 N, Eccentricity $e = 400$ mm, Rivet Spacing $C = 100$ mm, Rivet Count $n = 7$, Permissible Shear Stress $\tau = 65$ MPa, Permissible Crushing Stress $\sigma_c = 120$ MPa.

Design Protocol: Calculate center of gravity $G(x \mid, y \mid)$ of 7-rivet system, compute radial distances l_i , evaluate direct shear load P_s , compute secondary shear loads F_i from moment $M = Pe$, calculate resultant loads R_3, R_4, R_5 , size rivet hole diameter d for max load $R_5 = 33121$ N, and verify crushing stress σ_c .

1. Center of Gravity of Rivet Group (G) Coordinate centroid from origin O(0,0)	$x \mid = \frac{\sum x_i}{n} = \frac{0 + 100 + 200 + 200 + 200 + 100 + 0}{7} = \frac{700}{7} = \mathbf{100\text{ mm}}$ $y \mid = \frac{\sum y_i}{n} = \frac{200 + 200 + 200 + 100 + 0 + 0 + 0}{7} = \frac{800}{7} = \mathbf{114.3\text{ mm}}$
2. Radial Distances from Centroid (l_i) Radial distances of 7 rivets from $G(100, 114.3)$	$l_1 = l_3 = \sqrt{(100)^2 + (200 - 114.3)^2} = \sqrt{10000 + 7344.5} = \mathbf{131.7\text{ mm}}$ $l_2 = 200 - 114.3 = \mathbf{85.7\text{ mm}}$ $l_4 = l_7 = \sqrt{(100)^2 + (114.3 - 100)^2} = \sqrt{10000 + 204.5} = \mathbf{101\text{ mm}}$ $l_5 = l_6 = \sqrt{(100)^2 + (114.3)^2} = \sqrt{10000 + 13064.5} = \mathbf{152\text{ mm}}$
3. Direct Shear Load (P_s) Primary vertical shear load per rivet	$P_s = \frac{P}{n}$ $= \frac{50000}{7}$ $= \mathbf{7143\text{ N (Vertically Downward)}}$

4. Moment & Secondary Shear Loads ($M, \sum l_i^2, F_i$)

Turning moment about G and secondary shear load distribution

$$M = P \cdot e = 50000 \times 400 = 20 \times 10^6 \text{ N-mm}$$
$$\sum l_i^2 = 2(131.7)^2 + (85.7)^2 + 2(101)^2 + 2(152)^2 = \mathbf{108645\text{mm}^2}$$
$$F_1 = F_3 = \frac{M \cdot l_1}{\sum l_i^2} = \frac{20 \times 10^6 \times 131.7}{108645} = \mathbf{24244 \text{ N}}$$
$$F_2 = F_1 \left(\frac{l_2}{l_1} \right) = 24244 \left(\frac{85.7}{131.7} \right) = \mathbf{15776 \text{ N}}$$
$$F_4 = F_7 = F_1 \left(\frac{l_4}{l_1} \right) = 24244 \left(\frac{101}{131.7} \right) = \mathbf{18593 \text{ N}}$$
$$F_5 = F_6 = F_1 \left(\frac{l_5}{l_1} \right) = 24244 \left(\frac{152}{131.7} \right) = \mathbf{27981 \text{ N}}$$

5. Resultant Shear Loads on Critical Rivets (R_3, R_4, R_5)

Combining primary and secondary shear vector forces

$$R_3 = \sqrt{P_s^2 + F_3^2 + 2P_s F_3 \cos \theta_3} = \sqrt{7143^2 + 24244^2 + 2(7143)(24244)(\cos 90^\circ)} = \mathbf{25000 \text{ N}}$$
$$R_4 = \sqrt{P_s^2 + F_4^2 + 2P_s F_4 \cos \theta_4} = \sqrt{7143^2 + 18593^2 + 2(7143)(18593)(\cos 90^\circ)} = \mathbf{19300 \text{ N}}$$
$$R_5 = \sqrt{P_s^2 + F_5^2 + 2P_s F_5 \cos \theta_5} = \sqrt{7143^2 + 27981^2 + 2(7143)(27981)(\cos 90^\circ)}$$

= **33121 N** (Governing Maximum Load on Rivet 5)

6. Rivet Hole Diameter (d)

Single shear strength capacity for maximum load $R_5 = 33121 \text{ N}$

$$R_5 = \frac{\pi}{4} d^2 \cdot \tau$$
$$33121 = \frac{\pi}{4} d^2 (65) = 51.05 d^2$$
$$d^2 = \frac{33121}{51.05} = 648.8$$

⇒ **Standard Hole Diameter $d = 25.5 \text{ mm}$**

⇒ **Nominal Rivet Diameter $d_0 = 24 \text{ mm}$**

7. Crushing Stress Verification (σ_c)

Induced crushing stress check on critical rivet 5

$$\begin{aligned}\sigma_c &= \frac{R_5}{d \cdot t} \\ &= \frac{33121}{25.5 \times 25} \\ &= \frac{33121}{637.5} \\ &= 51.95 \text{ MPa} \quad (\text{Satisfactory, well below allowable } 120 \text{ MPa})\end{aligned}$$

Design Output Summary

7-Rivet Bracket t = 25 mm | Centroid G = (100, 114.3) | Direct Shear Ps = 7143 N |
Moment M = 20 MN-mm | Max Load R5 = 33121 N | Hole d = 25.5 mm | Rivet d0 = 24 mm |
Crushing Stress = 51.95 MPa

SECTION 15: SYMMETRIC 9-RIVET ECCENTRIC BRACKET

System Parameters: Eccentric Load $P = 45\text{ kN} = 45000\text{ N}$, Eccentricity $e = 500\text{ mm}$, Rivet Grid 3×3 ($n = 9$), Horizontal Pitch = 100 mm, Vertical Pitch = 120 mm, Permissible Shear Stress $\tau = 40\text{ MPa}$.

Design Protocol: Locate centroid G at center rivet 5, determine radial distances l_i , evaluate direct shear load $P_s = 5000\text{ N}$, compute secondary shear loads F_i from moment $M = Pe = 22.5 \times 10^6\text{ N-mm}$, determine maximum resultant shear load $R_3 = R_9 = 26600\text{ N}$, and size rivet hole diameter d .

1. Centroid & Radial Distances (G, l_i) Symmetric 3x3 rivet grid centered at rivet 5 $G(0, 0)$	$l_1 = l_3 = l_7 = l_9 = \sqrt{(100)^2 + (120)^2} = \sqrt{10000 + 14400} = \mathbf{156.2\text{ mm}}$$l_2 = l_8 = 120\text{ mm}, \quad l_4 = l_6 = 100\text{ mm}, \quad l_5 = 0$
2. Direct Shear Load (P_s) Primary vertical shear force acting on each rivet	$P_s = \frac{P}{n}$$= \frac{45000}{9}$$= \mathbf{5000\text{ N (Vertically Downward)}}$
3. Moment & Secondary Shear Loads ($M, \sum l_i^2, F_i$) Torsional moment about G and secondary shear load distribution	$M = P \cdot e = 45000 \times 500 = 22.5 \times 10^6\text{ N-mm}$$\sum l_i^2 = 4(156.2)^2 + 2(120)^2 + 2(100)^2 = \mathbf{146400\text{mm}^2}$$F_1 = F_3 = F_7 = F_9 = \frac{M \cdot l_1}{\sum l_i^2} = \frac{22.5 \times 10^6 \times 156.2}{146400} = \mathbf{23120\text{ N}}$$F_2 = F_8 = 23120 \left(\frac{120}{156.2} \right) = \mathbf{17762\text{ N}}$$F_4 = F_6 = 23120 \left(\frac{100}{156.2} \right) = \mathbf{14800\text{ N}}$

4. Resultant Shear Loads on Critical Rivets 3 & 9 (R_3, R_9)

Vector addition of direct and secondary shear forces ($\cos \theta_3 = \frac{100}{156.2} = 0.64$)

$$\begin{aligned} R_3 = R_9 &= \sqrt{P_s^2 + F_3^2 + 2P_s F_3 \cos \theta_3} \\ &= \sqrt{5000^2 + 23120^2 + 2(5000)(23120)(0.64)} \\ &= \sqrt{25 \times 10^6 + 534.5 \times 10^6 + 147.97 \times 10^6} \\ &= \mathbf{26600 \text{ N}} \text{ (Governing Maximum Resultant Load)} \\ R_6 &= P_s + F_6 = 5000 + 14800 = 19800 \text{ N} \end{aligned}$$

5. Rivet Hole Diameter (d)

Single shear strength capacity for maximum load $R_3 = 26600 \text{ N}$

$$\begin{aligned} R_3 &= \frac{\pi}{4} d^2 \cdot \tau \\ 26600 &= \frac{\pi}{4} d^2 (40) = 31.42 d^2 \\ d^2 &= \frac{26600}{31.42} = 846.6 \\ &\Rightarrow \text{Standard Hole Diameter } d = \mathbf{29 \text{ mm}} \\ &\Rightarrow \text{Nominal Rivet Diameter } d_0 = \mathbf{27 \text{ mm}} \end{aligned}$$

Design Output Summary

9-Rivet Bracket P = 45 kN, e = 500 mm | Direct Shear Ps = 5000 N | Moment M = 22.5 MN-mm | Max Resultant R3 = 26600 N | Hole d = 29 mm | Rivet d0 = 27 mm

SECTION 16: MAXIMUM SAFE ECCENTRIC LOAD CAPACITY

System Parameters: Number of Rivets $n = 4$ in horizontal line (A, B, C, D), Rivet Hole Diameter $d = 20$ mm, Allowable Shear Stress $\tau = 100$ MPa = 100N/mm², Eccentricity $e = 100$ mm, Spacings $l_A = l_D = 300$ mm, $l_B = l_C = 100$ mm.

Design Protocol: Express direct shear load $P_s = 0.25P$ and secondary shear loads (F_A, F_B, F_C, F_D) as functions of load P , determine resultant loads (R_A, R_B, R_C, R_D), equate maximum load $R_D = 0.40P$ to single rivet shear capacity (31420 N), and calculate maximum safe eccentric load P .

1. Direct Shear Load per Rivet (P_s) Primary vertical shear load per rivet in terms of load P	$P_s = \frac{P}{n} = \frac{P}{4} = \mathbf{0.25P\ N\ (Vertically\ Upward)}$
2. Moment & Secondary Shear Loads ($M, \sum l_i^2, F_i$) Torsional moment about centroid E and secondary load distribution	$M = P \cdot e = P \times 100 = \mathbf{100P\ N\text{-}mm\ (Anticlockwise)}$$\sum l_i^2 = 2(300)^2 + 2(100)^2 = 180000 + 20000 = \mathbf{200000mm^2}$$F_A = F_D = \frac{M \cdot l_A}{\sum l_i^2} = \frac{100P \times 300}{200000} = \mathbf{0.15P\ N}$$F_B = F_C = \frac{M \cdot l_B}{\sum l_i^2} = \frac{100P \times 100}{200000} = \mathbf{0.05P\ N}$
3. Resultant Shear Load on Each Rivet (R_A, R_B, R_C, R_D) Vector sum of direct and secondary shear loads	$R_A = P_s - F_A = 0.25P - 0.15P = 0.10P$$R_B = P_s - F_B = 0.25P - 0.05P = 0.20P$$R_C = P_s + F_C = 0.25P + 0.05P = 0.30P$$R_D = P_s + F_D = 0.25P + 0.15P = \mathbf{0.40P\ N\ (Governing\ Maximum)}$

4. Single Rivet Shear Capacity (P_{shear})

Maximum allowable shear force capacity per rivet ($d = 20 \text{ mm}$)

$$\begin{aligned} P_{\text{shear}} &= \frac{\pi}{4} d^2 \cdot \tau \\ &= \frac{\pi}{4} (20)^2 \times 100 \\ &= 100\pi \times 100 \\ &= \mathbf{31420 \text{ N} = 31.42 \text{ kN}} \end{aligned}$$

5. Maximum Safe Eccentric Load (P)

Equating governing load $R_D = 0.40P$ to single rivet shear capacity

$$\begin{aligned} 0.40P &= 31420 \text{ N} \\ P &= \frac{31420}{0.40} \\ &= 78550 \text{ N} \\ \Rightarrow \text{Maximum Safe Load } P &= \mathbf{78.55 \text{ kN}} \end{aligned}$$

Design Output Summary

4-Rivet Horizontal Joint d = 20 mm | Eccentricity e = 100 mm | Governing Load RD = 0.40 P | Single Rivet Capacity = 31.42 kN | Max Safe Load P = 78.55 kN

SECTION 17: ECCENTRIC COLUMN BRACKET JOINT

System Parameters: Number of Rivets $n = 6$ (2 columns of 3 rivets), Eccentric Load $P = 60 \text{ kN} = 60000 \text{ N}$, Eccentricity $e = 200 \text{ mm}$, Horizontal Spacing = 150 mm (75 mm from G), Vertical Spacing = 100 mm (50 mm from G), Permissible Shear Stress $\tau = 150 \text{ MPa}$.

Design Protocol: Locate symmetric centroid $G(0, 0)$, compute radial distances l_i , evaluate direct shear load $P_s = 10000 \text{ N}$, compute secondary shear loads F_i from moment $M = Pe = 12 \times 10^6 \text{ N-mm}$, determine maximum resultant shear load $R_1 = R_3 = 35384 \text{ N}$, and size rivet hole diameter d .

1. Centroid & Radial Distances (G, l_i)

Symmetric 2x3 rivet grid centered at centroid $G(0, 0)$

$l_1 = l_3 = l_4 = l_6 = \sqrt{(75)^2 + (50)^2} = \sqrt{5625 + 2500} = \mathbf{90.1 \text{ mm}}$

$l_2 = l_5 = 50 \text{ mm}$

2. Direct Shear Load (P_s)

Primary vertical shear force acting on each rivet

$P_s = \frac{P}{n}$

$= \frac{60000}{6}$

$= \mathbf{10000 \text{ N (Vertically Downward)}}$

3. Moment & Secondary Shear Loads ($M, \sum l_i^2, F_i$)

Torsional moment about G and secondary shear load distribution

$M = P \cdot e = 60000 \times 200 = 12 \times 10^6 \text{ N-mm}$

$\sum l_i^2 = 4(90.1)^2 + 2(50)^2 = 32500 + 5000 = \mathbf{37500\text{mm}^2}$

$F_1 = F_3 = F_4 = F_6 = \frac{M \cdot l_1}{\sum l_i^2} = \frac{12 \times 10^6 \times 90.1}{37500} = \mathbf{28846 \text{ N}}$

$F_2 = F_5 = 28846 \left(\frac{50}{90.1}\right) = \mathbf{16008 \text{ N}}$

4. Resultant Shear Loads on Critical Rivets 1 & 3 (R_1, R_3)

Vector addition of direct and secondary shear forces ($\cos \theta_1 = \frac{50}{90.1} = 0.555$)

$$\begin{aligned} R_1 = R_3 &= \sqrt{P_s^2 + F_1^2 + 2P_sF_1 \cos \theta_1} \\ &= \sqrt{10000^2 + 28846^2 + 2(10000)(28846)(0.555)} \\ &= \sqrt{100 \times 10^6 + 832.1 \times 10^6 + 320.2 \times 10^6} \\ &= \mathbf{35384 \text{ N (Governing Maximum Resultant Load)}} \\ R_2 &= P_s + F_2 = 10000 + 16008 = 26008 \text{ N} \end{aligned}$$

5. Rivet Hole Diameter (d)

Single shear strength capacity for maximum load $R_1 = 35384 \text{ N}$

$$\begin{aligned} R_1 &= \frac{\pi}{4} d^2 \cdot \tau \\ 35384 &= \frac{\pi}{4} d^2 (150) = 117.8 d^2 \\ d^2 &= \frac{35384}{117.8} = 300.4 \\ &\Rightarrow \mathbf{\text{Standard Hole Diameter } d = 19.5 \text{ mm}} \\ &\Rightarrow \mathbf{\text{Nominal Rivet Diameter } d_0 = 18 \text{ mm}} \end{aligned}$$

Design Output Summary

6-Rivet Column Bracket P = 60 kN, e = 200 mm | Direct Shear Ps = 10000 N | Moment M = 12 MN-mm | Max Resultant R1 = 35384 N | Hole d = 19.5 mm | Rivet d0 = 18 mm

SECTION 18: ECCENTRICALLY LOADED INCLINED BRACKET JOINT

System Parameters: Number of Rivets $n = 4$ (A, B, C, D in vertical line), Spacings $l_A = l_D = 90$ mm, $l_B = l_C = 30$ mm, Inclined Load $P = 100$ kN = 100000 N at 30^deg to horizontal, Arm EF = 150 mm, Yield Shear Stress $\tau_y = 240$ MPa, FOS = 1.5, Allowable Bending Stress $\sigma_b = 125$ MPa, Plate Width $b = 240$ mm.

Design Protocol: Calculate direct shear load $P_s = 25000$ N, perpendicular eccentricity $e = 75$ mm, moment $M = 7.5 \times 10^6$ N-mm, secondary shear loads F_i , governing resultant load $R_D = 60455$ N, size rivet hole $d = 23.5$ mm (nominal $d_0 = 22$ mm), compute weakest section moment of inertia $I_{xx} = 724674t$, and solve bracket plate thickness $t = 10$ mm.

1. Direct Shear Load (P_s)
Primary shear load per rivet at 30^deg angle to horizontal

$$P_s = \frac{P}{n}$$
$$= \frac{100000}{4}$$
$$= \mathbf{25000 \text{ N (at 30 deg to horizontal)}}$$

2. Perpendicular Eccentricity & Moment (e, M)
Perpendicular moment arm from E to load line of action

$$e = \text{EF} \sin(30^\text{deg}) = 150 \times 0.5 = \mathbf{75 \text{ mm}}$$
$$M = P \cdot e = 100000 \times 75 = \mathbf{7.5 \times 10^6 \text{ N-mm}}$$

3. Secondary Shear Loads (F_A, F_B, F_C, F_D)
Torsional shear distribution ($\sum l_i^2 = 2(90)^2 + 2(30)^2 = 18000\text{mm}^2$)

$$F_A = F_D = \frac{M \cdot l_A}{\sum l_i^2} = \frac{7.5 \times 10^6 \times 90}{18000} = \mathbf{37500 \text{ N}}$$
$$F_B = F_C = \frac{M \cdot l_B}{\sum l_i^2} = \frac{7.5 \times 10^6 \times 30}{18000} = \mathbf{12500 \text{ N}}$$

4. Resultant Shear Loads (R_A, R_B, R_C, R_D)

Vector addition of primary (P_s) and secondary (F_i) shear loads

$$\begin{aligned} R_A &= R_B = \sqrt{P_s^2 + F_A^2 + 2P_sF_A \cos(150^\circ)} = \mathbf{19365\text{ N}} \\ R_C &= \sqrt{P_s^2 + F_C^2 + 2P_sF_C \cos(30^\circ)} = \mathbf{36366\text{ N}} \\ R_D &= \sqrt{P_s^2 + F_D^2 + 2P_sF_D \cos(30^\circ)} \\ &= \sqrt{(25000)^2 + (37500)^2 + 2(25000)(37500)(0.866)} \\ &= \mathbf{60455\text{ N (Governing Maximum Resultant Load on Rivet D)}} \end{aligned}$$

5. Rivet Hole Diameter (d)

Permissible shear stress $\tau_{\text{allow}} = \frac{240}{1.5} = 160\text{ MPa}$ for load $R_D = 60455\text{ N}$

$$\begin{aligned} R_D &= \frac{\pi}{4}d^2 \cdot \left(\frac{\tau_y}{\text{FOS}} \right) \\ 60455 &= \frac{\pi}{4}d^2(160) = 125.7d^2 \\ d^2 &= \frac{60455}{125.7} = 481 \\ \Rightarrow \text{Standard Hole Diameter } d &= \mathbf{23.5\text{ mm}} \\ \Rightarrow \text{Nominal Rivet Diameter } d_0 &= \mathbf{22\text{ mm}} \end{aligned}$$

6. Moment of Inertia of Weakest Section (I_{xx})

Net area moment of inertia deducting 4 rivet holes ($d = 23.5\text{ mm}$)

$$\begin{aligned} I_{xx} &= \frac{tb^3}{12} - \left[4\frac{td^3}{12} + 2(t \cdot d)(90)^2 + 2(t \cdot d)(30)^2 \right] \\ &= \frac{t(240)^3}{12} - [4326t + 2(23.5t)(8100 + 900)] \\ &= 1152000t - [4326t + 423000t] \\ &= \mathbf{724674t\text{mm}^4} \end{aligned}$$

7. Bracket Plate Thickness (*t*)

Bending stress formula for allowable stress $\sigma_b = 125 \text{ MPa}$ ($y = \frac{b}{2} = 120 \text{ mm}$)

$$\sigma_b = \frac{M \cdot y}{I_{xx}}$$
$$125 = \frac{7.5 \times 10^6 \times 120}{724674t}$$
$$125 = \frac{1241.9}{t}$$
$$t = \frac{1241.9}{125} = 9.935 \text{ mm}$$

$\Rightarrow$ Adopt Bracket Plate Thickness $t = 10 \text{ mm}$

Design Output Summary

Inclined Load P = 100 kN at 30 deg | Moment M = 7.5 MN-mm | Max Load RD = 60.46 kN | Hole d = 23.5 mm | Rivet d0 = 22 mm | Weakest Section Ixx = 724674 t | Plate Thickness t = 10 mm